An exact finite-period Fourier–SOC criterion for primal-first Chambolle–Pock cycles

Abstract

For every integer $m \geq 2$, every real extrapolation parameter, and every positive actual stepsize product, we characterize the existence of nonconstant m-periodic orbits of the literal primal-first Chambolle–Pock map by a finite second-order-cone system. Necessity holds for arbitrary finite-dimensional real Hilbert spaces, linear couplings, and proper closed convex functions. Every feasible point is realized by two convex-hull indicators in dimension $2(m-1)$, with the prescribed product, a saddle pair, and literal paired initialization. All $2(m-1)$ cyclic inequalities are retained, including the real Nyquist mode and degenerate Gram data. Least period is characterized by the greatest common divisor of the combined primal and dual frequency support; its strict support conditions are distinguished from closed conic feasibility. We prove compactness, positive product bounds, product-ray properties, and exact scalar consequences for period two and zero extrapolation. These are finite-cycle results, not global convergence assertions. No priority or originality claim is made.

Keywords: Chambolle–Pock iteration; finite-period orbits; Fourier decomposition; second-order-cone feasibility; convex interpolation; least-period criterion

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1 Recurrence and scope

Let X, Y be finite-dimensional real Hilbert spaces, $L : X \to Y$ linear, and f, h proper closed convex extended-real functions. For $a > 0$, write

$$\operatorname{prox}_{af}(z) = \arg\min_x \{f(x) + \|x - z\|^2/(2a)\}.$$

Fix $\theta \in \mathbb{R}$ and $\tau, \sigma > 0$. The map under study is exactly

$$\begin{aligned} x_{i+1} &= \operatorname{prox}_{\tau f}(x_i - \tau L^* y_i), \\ y_{i+1} &= \operatorname{prox}_{\sigma h}\big(y_i + \sigma L((1+\theta)x_{i+1} - \theta x_i)\big). \end{aligned} \tag{1}$$

Its state is the pair (x_i, y_i); no extrapolated history is initialized. The product is the *actual* value $r = \tau\sigma\|L\|^2$, not a chosen bound for it. A saddle pair means $-L^* y_* \in \partial f(x_*)$ and $Lx_* \in \partial h(y_*)$.

2 The complete conic system

Fix an integer $m \geq 2$ and $r > 0$. Put

$$I_m = \{1, \ldots, \lfloor m/2 \rfloor\}, \qquad I_m^{\mathrm{c}} = \{1, \ldots, \lfloor (m-1)/2 \rfloor\}, \qquad \phi_k = 2\pi k/m.$$

For each $k \in I_m$ introduce real variables A_k, B_k, c_k; introduce d_k only for $k \in I_m^{\mathrm{c}}$. If m is even, set $d_{m/2} = 0$. For every $j = 1, \ldots, m-1$, define

$$\begin{aligned} D_{kj} &= \cos((j+1)\phi_k) - \cos\phi_k - \cos(j\phi_k) + 1, \\ Z_{kj} &= e^{\mathrm{i}\phi_k}(e^{\mathrm{i}j\phi_k} - 1), \\ W_{kj} &= ((1+\theta) - \theta e^{\mathrm{i}\phi_k})(e^{\mathrm{i}j\phi_k} - 1). \end{aligned} \tag{2}$$

The system consists of

$$A_k, B_k \geq 0, \qquad c_k^2 + d_k^2 \leq A_k B_k, \qquad \sum_{k \in I_m} A_k = 1, \tag{3}$$

and *all* $2(m-1)$ inequalities

$$\begin{aligned} P_j &= \sum_{k \in I_m} \left(\frac{D_{kj}}{r} A_k - (\operatorname{Re} Z_{kj}) c_k - (\operatorname{Im} Z_{kj}) d_k \right) \leq 0, \\ H_j &= \sum_{k \in I_m} \left(D_{kj} B_k + (\operatorname{Re} W_{kj}) c_k - (\operatorname{Im} W_{kj}) d_k \right) \leq 0. \end{aligned} \tag{4}$$

For fixed m, θ, r, this is a second-order-cone (SOC) feasibility problem: the quadratic constraint is equivalently

$$\|(2c_k, 2d_k, A_k - B_k)\|_2 \leq A_k + B_k,$$

with the $2d_k$ coordinate omitted at Nyquist. All other constraints are linear. If m is even, put $e_j = 1 - (-1)^j$. The Nyquist contributions to P_j, H_j are, respectively,

$$e_j \left(\frac{2A_{m/2}}{r} - c_{m/2} \right), \qquad e_j \left(2B_{m/2} - (1+2\theta) c_{m/2} \right). \tag{5}$$

Thus the real mode has no fictitious imaginary degree of freedom.

Theorem 1. *For every integer $m \geq 2$, every $\theta \in \mathbb{R}$, and every $r > 0$, the following statements are equivalent:*

1. *Some map* (1) *with actual product r has a nonconstant paired orbit with period dividing m.*
2. *Such an orbit exists for a map that also has a saddle pair.*
3. *The full system* (3)–(4) *is feasible.*

Every feasible point is realized by convex-hull indicators with $X = Y = \mathbb{R}^{2(m-1)}$, $L = I$, $\tau = r$, $\sigma = 1$, and saddle pair $(0, 0)$. The constructed orbit starts from its literal pair (x_0, y_0). The dimension is an upper bound, not a minimality assertion.

Theorem 2. *Replace "nonconstant paired orbit with period dividing m" in Theorem 1 by "paired orbit of least period m." The equivalence remains true if statement 3 also imposes*

$$\gcd\big(m, \{k \in I_m : A_k + B_k > 0\}\big) = 1. \tag{6}$$

The same realization and saddle-pair assertions hold.

The support in (6) is the union of the primal and dual supports. In particular, $A_k = 0$ forces $c_k = d_k = 0$, but need not force $B_k = 0$. For prime m, every nonempty support has gcd one, so the two existence questions coincide.

The extra condition is a finite union of relatively open conditions: for at least one subset $S \subseteq I_m$ with $\gcd(m, S) = 1$, require $A_k + B_k > 0$ for all $k \in S$. Modes outside S may be present. Thus Theorem 2 is not, in general, a claim of ordinary closed-SOC feasibility.

3 Necessity: the actual map and every cyclic lag

Suppose a paired orbit is m-periodic; all indices in this proof are modulo m. Each state is a proximal output, including state zero, since the last transition returns to the initial pair. Its function values are finite, and optimality gives

$$\begin{aligned} a_i &= (x_{i-1} - x_i)/\tau - L^* y_{i-1} \in \partial f(x_i), \\ b_i &= (y_{i-1} - y_i)/\sigma + L((1+\theta)x_i - \theta x_{i-1}) \in \partial h(y_i). \end{aligned} \tag{7}$$

For *every* displacement $j = 1, \dots, m-1$, sum the subgradient inequalities over i. Function values cancel, yielding

$$\sum_i \langle a_i, x_{i+j} - x_i \rangle \le 0, \qquad \sum_i \langle b_i, y_{i+j} - y_i \rangle \le 0. \tag{8}$$

This step uses convex interpolation inequalities, not merely pairwise monotonicity.

Set $U_i = (x_i - \bar{x})/\sqrt{\tau}$, $V_i = (y_i - \bar{y})/\sqrt{\sigma}$, and $K = \sqrt{\tau\sigma}L$. Equip temporal sequences with the inner product $m^{-1}\sum_i \langle \cdot, \cdot \rangle$. Let F be the forward cyclic shift and $S = F^{-1}$. Lift K diagonally to temporal sequences. It commutes with shifts, preserves their frequency spaces, and has exactly the original norm $\sqrt{r}$. Constants disappear from (8), which becomes

$$\begin{aligned} \langle (S - I)U - K^* S V, (F^j - I)U \rangle \le 0, \\ \langle (S - I)V + K((1+\theta)I - \theta S)U, (F^j - I)V \rangle \le 0. \end{aligned} \tag{9}$$

The centered real temporal space is the orthogonal sum of frequency spaces indexed by I_m. On each complex mode $k \in I_m^{\mathrm{c}}$,

$$F = \cos\phi_k I + \sin\phi_k J_k, \qquad J_k^* = -J_k, \qquad J_k^2 = -I.$$

Here K intertwines the corresponding structures on the primal and dual spaces. At even-m Nyquist, $F = S = -I$ and there is no complex structure. Write U_k, V_k for frequency projections and set

$$A_k = r\|U_k\|^2, \quad B_k = \|V_k\|^2, \quad c_k = \langle KU_k, V_k\rangle, \quad d_k = \langle KJ_kU_k, V_k\rangle. \tag{10}$$

At Nyquist define only A_k, B_k, c_k and put $d_k = 0$.

In a complex mode, KU_k and $KJ_kU_k = J_kKU_k$ are orthogonal and have equal norm. Bessel's inequality, followed by the actual norm bound, gives

$$c_k^2 + d_k^2 \le \|KU_k\|^2\|V_k\|^2 \le A_kB_k. \tag{11}$$

If $KU_k = 0$, the same conclusion follows directly. At Nyquist ordinary Cauchy–Schwarz gives $c_k^2 \le A_kB_k$. No spatial rank assumption is involved.

Orthogonality splits (9) by frequency. The self term has scalar coefficient D_{kj}, because the symmetric part of $(F - I)(F^j - I)$ is $D_{kj}I$. The primal cross term is

$$-\langle V_k, KF(F^j - I)U_k\rangle = -(\operatorname{Re} Z_{kj})c_k - (\operatorname{Im} Z_{kj})d_k.$$

For the dual cross term, move $F^j - I$ to its adjoint. The resulting multiplier on U_k is

$$(F^{-j} - I)((1 + \theta)I - \theta F^{-1}),$$

whose complex scalar is $\overline{W_{kj}}$. Its contribution is $(\operatorname{Re} W_{kj})c_k - (\operatorname{Im} W_{kj})d_k$. Substitution of $F = -I$ gives (5). Thus (9) is exactly (4) before normalization.

If $U = 0$, the dual inequality at $j = m - 1$ reduces to $\|(S - I)V\|^2 \le 0$. Centering then gives $V = 0$. Hence a nonconstant paired orbit satisfies $\sum_k A_k > 0$. Divide A, B, c, d by this positive sum. All cones and cyclic inequalities are homogeneous, giving precisely (3)–(4).

Finally, a centered periodic sequence has period p exactly when F^p fixes it. A nonzero component at frequency k is fixed exactly when m divides kp, including Nyquist. Therefore the least period of the *paired* sequence is

$$\frac{m}{\gcd\bigl(m, \{k : \|U_k\|^2 + \|V_k\|^2 > 0\}\bigr)}. \tag{12}$$

Centering and positive rescaling preserve support. This proves necessity for both theorems, including the use of $A_k + B_k$ rather than A_k alone.

4 Sufficiency: explicit indicators and all normal tests

Take any feasible point. For each complex mode use two orthogonal real planes, on both of which T rotates through ϕ_k. For even m, add two axes on which $T = -I$. Make all blocks mutually orthogonal. The total dimension is

$$4\lfloor (m-1)/2 \rfloor + 2\,\mathbf{1}_{\{m \text{ even}\}} = 2(m-1).$$

We have $T^m = I$ and $\sum_{i=0}^{m-1} T^i = 0$. For a complex mode with $A_k > 0$, use coordinates in its two planes to set

$$\begin{aligned} x_{0,k} &= (\sqrt{A_k}, 0\,;\, 0, 0), \\ y_{0,k} &= \left(\frac{c_k}{\sqrt{A_k}}, \frac{d_k}{\sqrt{A_k}}\,;\, \sqrt{B_k - \frac{c_k^2 + d_k^2}{A_k}}, 0 \right). \end{aligned} \tag{13}$$

If $A_k = 0$, use $x_{0,k} = 0$ and $y_{0,k} = (0, 0\,;\, \sqrt{B_k}, 0)$. The cone makes $c_k = d_k = 0$, so this retains every possible dual-only component. At Nyquist, if $A > 0$, use

$$x_0 = (\sqrt{A}, 0), \qquad y_0 = (c/\sqrt{A}, \sqrt{B - c^2/A});$$

if $A = 0$, use $x_0 = 0$ and $y_0 = (0, \sqrt{B})$. All square roots are well-defined by the cones.

Combine the blocks into x_0, y_0, and define

$$\begin{aligned} &x_i = T^i x_0, \qquad y_i = T^i y_0, \\ &C = \operatorname{conv}\{x_0, \dots, x_{m-1}\}, \qquad D = \operatorname{conv}\{y_0, \dots, y_{m-1}\}, \\ &f = \iota_C, \qquad h = \iota_D, \qquad L = I, \qquad \tau = r, \qquad \sigma = 1. \end{aligned} \tag{14}$$

The hulls are nonempty compact convex sets; their indicators are proper closed convex. The actual norm of L is one even if the orbit occupies a proper subspace, so the actual product is exactly r.

Set

$$a_i = (x_{i-1} - x_i)/r - y_{i-1}, \qquad b_i = y_{i-1} - y_i + (1+\theta)x_i - \theta x_{i-1}.$$

For the blocks above, the Gram data are exactly

$$A_k = \|x_{0,k}\|^2, \quad B_k = \|y_{0,k}\|^2, \quad c_k = \langle x_{0,k}, y_{0,k} \rangle, \quad d_k = \langle J_k x_{0,k}, y_{0,k} \rangle.$$

The coefficient calculation already proved gives, for every $j = 1, \dots, m-1$,

$$\langle a_0, x_j - x_0 \rangle = P_j \le 0, \qquad \langle b_0, y_j - y_0 \rangle = H_j \le 0. \tag{15}$$

Equivariance yields $a_i = T^i a_0$ and $b_i = T^i b_0$. Orthogonality of T therefore transfers (15) to every vertex i and every displacement j. These are every off-diagonal comparison; the

diagonal comparison is zero, and repeated vertices cause no difficulty. Linearity extends each comparison to every point of the relevant hull. Thus

$$a_i \in N_C(x_i), \qquad b_i \in N_D(y_i)$$

for all i. These inclusions are exactly both proximal transitions in (1). Since proximal maps are single-valued, the actual run initialized at the literal pair (x_0, y_0) is the displayed sequence.

Normalization gives $\|x_0\|^2 = 1$, and there is no constant temporal mode, so the paired orbit is nonconstant. Formula (12) gives its least period with support $\{k : A_k + B_k > 0\}$. In particular (6) gives least period m.

The zero centroid identities give $0 \in C \cap D$. For a closed convex set, the zero vector belongs to its normal cone at every point of the set. Hence $-L^*0 \in \partial f(0)$ and $L0 \in \partial h(0)$: $(0, 0)$ is a saddle pair. Saddle-restricted existence plainly implies unrestricted existence. This completes the proofs of Theorems 1 and 2.

5 Compactness, product rays, and strict support tests

Put

$$s_m = 2\sin(\pi/m) > 0, \qquad M_\theta = |1 + \theta| + |\theta| \geq 1.$$

Proposition 5.1. At fixed $m \geq 2$, $\theta \in \mathbb{R}$, and $r > 0$, the normalized closed system has a compact feasible set, possibly empty. Every feasible point satisfies

$$\sum_k B_k \leq \frac{M_\theta^2}{s_m^2}, \qquad r \geq \frac{s_m^2}{M_\theta}. \tag{16}$$

At fixed m, the dividing-period feasible parameter set is relatively closed in $\mathbb{R} \times (0, \infty)$. At fixed θ, if nonempty, its feasible products form an attained upper ray $[r_{\mathrm{div}}, \infty)$.

Proof. Let $\delta_k = |e^{\mathrm{i}\phi_k} - 1| \geq s_m$ and $E = \sum_k \delta_k^2 B_k$. At the last lag $j = m - 1$, we have $D_{kj} = \delta_k^2$ and $|W_{kj}| \leq M_\theta \delta_k$. Using the cones and Cauchy–Schwarz,

$$0 \geq H_{m-1} \geq E - M_\theta \sum_k \delta_k \sqrt{A_k B_k} \geq E - M_\theta \sqrt{E},$$

since $\sum A_k = 1$. Thus $E \leq M_\theta^2$, proving the first bound. It bounds B; normalization bounds A; the cones bound c, d. All constraints are closed, proving compactness.

For the primal inequality at the same lag, $|Z_{kj}| = \delta_k$, so

$$\frac{s_m^2}{r} \leq \frac{\sum_k \delta_k^2 A_k}{r} \leq \sum_k \delta_k \sqrt{A_k B_k} \leq \sqrt{E} \leq M_\theta.$$

This proves the positive product bound. Along a convergent parameter sequence with limiting $r > 0$ and finite θ, these bounds are locally uniform. A convergent subsequence of witnesses passes through the continuous closed constraints, proving relative closedness.

For $r' \geq r$, let $\lambda = r/r' \in (0, 1]$ and set

$$A'_k = A_k, \qquad (B'_k, c'_k, d'_k) = \lambda(B_k, c_k, d_k). \tag{17}$$

The new cones hold because $\lambda^2(c_k^2 + d_k^2) \leq \lambda A_k B_k$. Each new expression P_j, H_j is exactly λ times the old one. Thus feasibility is upward closed. For a nonempty ray its infimum is positive by (16), is finite because the ray is nonempty, and is attained by relative closedness. This proves the assertion. ∎

Scaling (17) preserves the entire union support. Least-period-m products are therefore also upward closed. However, limiting witnesses can lose modes. Compactness of the containing closed system does *not* prove endpoint attainment under the strict support requirement, and no general least-period attainment assertion is made here.

There is nevertheless an exact compact optimization formulation of each strict test. For any $S \subseteq I_m$ with $\gcd(m, S) = 1$, adjoin a scalar η and constraints

$$0 \leq \eta \leq 1, \qquad A_k + B_k \geq \eta \quad (k \in S). \tag{18}$$

If the underlying closed system is nonempty, the augmented set is nonempty (take $\eta = 0$) and compact. Its maximum $\eta_{\max}$ is attained. A witness with every mode in S active gives positive feasible η by taking the minimum of one and its finitely many positive weights. Conversely $\eta_{\max} > 0$ supplies such a witness. Thus least-period feasibility is exactly the finite disjunction of tests $\eta_{\max} > 0$ over these subsets S; an empty closed system fails all tests.

If $r = 0$, then $L = 0$. The two proximal sequences decouple. Applying the last-lag inequality separately to each gives $\|(S - I)U\|^2 \leq 0$ and $\|(S - I)V\|^2 \leq 0$, so every finite-period paired orbit is constant. This case is separate from the conic system, which uses $1/r$.

6 Two exact scalar consequences

Corollary 6.1. *For every real θ and $r > 0$, a least-period-two orbit exists if and only if*

$$\theta > -\tfrac{1}{2}, \qquad r \geq \frac{4}{1 + 2\theta}.$$

The endpoint is attained and a saddle-pair realization exists.

Proof. Only the Nyquist mode remains, and normalization gives $A = 1$. The conditions are

$$4/r - 2c \leq 0, \qquad 4B - 2(1 + 2\theta)c \leq 0, \qquad c^2 \leq B.$$

Thus $c \geq 2/r > 0$ and $2c^2 \leq (1+2\theta)c$, so $c \leq (1+2\theta)/2$. This forces exactly the displayed necessary conditions. Conversely take $c = 2/r$ and $B = c^2$. All conditions hold, and the sole support index is one. Apply Theorem 2. ∎

Corollary 6.2. *At $\theta = 0$, for every integer $m \geq 2$ and $r > 0$, the following are equivalent: a nonconstant orbit with period dividing m exists; a least-period-m orbit exists; and*

$$r \geq 4\sin^2(\pi/m). \tag{19}$$

Saddle-pair realizations exist throughout this range.

Proof. Since $M_0 = 1$, (16) proves necessity. At equality put $\phi = 2\pi/m$ and $r = 2-2\cos\phi$. Activate only mode one with

$$A_1 = 1, \qquad B_1 = 1/r, \qquad c_1 = 1/2, \qquad d_1 = -\sin\phi/r$$

for $m > 2$, and $d_1 = 0$ for $m = 2$. The cone is saturated, since $1/4 + \sin^2\phi/r^2 = 1/r$.

To verify every displacement, let T be a rotation through ϕ (or $T = -I$ when $m = 2$), choose a unit x_0, and put $y_0 = (I - T)x_0/r$. The identity $T + T^{-1} = (2 - r)I$ gives

$$(T^{-1} - I)x_0/r - T^{-1}y_0 = 0, \qquad T^{-1}y_0 - y_0 + x_0 = 0.$$

Hence both residuals vanish and all P_j, H_j are zero. The support is $\{1\}$, so its least period is m. Scaling (17) supplies every larger product while preserving support. Theorem 2 gives the claimed saddle realizations. ∎

Remark. The existence assertion for every larger product in Corollary 6.2 allows higher-dimensional realizations. Product extension need not stay in a fixed planar regular-polygon ansatz. For example, starting with the saturated mode-one Gram data at $m > 2$, scaling by $0 < \lambda < 1$ gives the positive Gram determinant

$$A_1'B_1' - (c_1'^2 + d_1'^2) = \lambda(1 - \lambda)A_1B_1 > 0.$$

The two-plane realization accommodates this extra rank. Consequently this result does not contradict the restricted $\mathbb{R}^2$ regular fixed-ansatz classification in report 20. The two statements quantify over different classes of realizations.

7 Verification record and limitations

The proof above applies to arbitrary m; finite sampling is not its basis. The accompanying portable script `verify_exact.py` checks generic primal and dual coefficient identities symbolically with arbitrary rotation and displacement entries. It also verifies both Nyquist parities, last-lag identities, support-preserving product scaling, and the zero-extrapolation residual identities. Its exact symbolic equalities are algebraic cross-checks

of the proof, not a replacement for the Hilbert-space reduction, the realization argument, or the compactness proof.

A separate optional script `verify_samples.py` checks finitely many numerical realizations and finitely many integer supports. These finite checks are explicitly corroborative and do not establish a universal theorem. The accepted candidate received a complete independent arbitrary-period analytic audit before comparison with its proof; no material correction was required. The source package records reproducible verification commands and their outputs.

The result characterizes existence of finite cycles across the stated class of convex problems. It does not establish global convergence below any bound, describe all nonperiodic dynamics, claim a sharp global convergence boundary, or infer endpoint attainment for general least-period constraints. No claim concerning priority or originality is made.